

MODULE 3 OF 5 • PRINTABLE NOTES

Knowledge Organisation and Mind Mapping

Organise research information, synthesise source-based knowledge and transform research concepts into structured mind maps and literature matrices.

Research Matrix

Mind Mapping

Gemini Notebook Studio

Concept Mapping

🕒 LEARNING OBJECTIVES

- Generate a ready-to-use prompt for organising sources into a synthesis matrix, timeline or mind-map outline.
- Choose the right research matrix type — literature review, research design, or synthesis — for a given task.
- Extract concepts and relationships from an abstract and compare a manual map against an AI-generated one.
- Select an appropriate mind-mapping or knowledge-organisation tool for the task at hand.
- Keep every theme, cluster or connection traceable to a real source or note.

✓ BEFORE YOU START

- Bring 3 to 5 sources, notes or papers you want to organise, either from Module 1's search or your own shortlist.
- Have a free account ready for at least one mind-mapping tool (e.g., Xmind, MindMeister or Markmap).
- If using Gemini Notebook Studio, upload your verified sources before the session to save time.



KEY CONCEPTS

Glossary of terms

Terms you will meet throughout this module, defined in plain language.

Synthesis matrix

A table organising sources by theme, showing which studies support or address each theme.

Literature matrix

A structured table comparing sources across dimensions such as method, sample and findings.

Node

A single concept or entry in a mind map, connected to related ideas by branches.

Knowledge graph

A network of linked notes, sources or concepts showing how they relate.



Trainer's Tip, Dr. Shahizan: A mind map is only useful if you can still explain it a week later. Before you move on, talk through your map out loud to a labmate. If you cannot explain a link between two nodes, it probably needs a source or it needs to go.



AI TOOLS FOR THIS MODULE

Mind mapping and knowledge organisation tools

TOOL	CATEGORY	WHAT IT DOES	WHERE TO ACCESS
Xmind	Mind mapping	Structured mind maps with outline, tree and fishbone diagram views.	xmind.app
Markmap	Markdown → mind map	Transforms Markdown outlines into interactive mind maps.	markmap.js.org
MindMeister	Collaborative mind mapping	Online collaborative mind-mapping tool for concepts and projects.	mindmeister.com
Obsidian	Networked notes	Links notes, sources and concepts into a navigable knowledge graph.	obsidian.md
Gemini Notebook Studio	Source-grounded	Generates a mind map directly from your uploaded, verified sources.	notebooklm.google.com
Notion	All-in-one workspace	Database views and linked pages for a running literature matrix.	notion.so
Miro	Visual whiteboard	Collaborative whiteboard for clustering literature notes and framework diagrams.	miro.com
Whimsical	Mind maps & flowcharts	Fast, lightweight mind maps and wireframes for early-stage structuring.	whimsical.com



HOW TO USE

Step-by-step tool guides



STEP-BY-STEP GUIDE

Knowledge Organisation Prompt Builder

Describe the sources or notes you want to organise. The builder assembles a ready-to-use prompt asking an AI chat tool to structure them into a synthesis matrix, timeline, mind map outline or theoretical framework map.

01

Describe

List the sources/notes or describe the topic and concepts you want structured.

02

Choose

Pick an organisation goal (thematic matrix, timeline, mind map, framework map) and output format (table, outline, narrative).

03

Ground it

Upload the actual papers or notes to the AI tool alongside the prompt so every theme is traceable to a real source.



STEP-BY-STEP GUIDE

Research Matrix Builder

A research matrix forces structure onto messy source material or study design — one row per source, question or theme — so patterns, gaps and inconsistencies become visible at a glance. Pick the type that matches what you need right now.

01

Literature Review Matrix

Organise past studies by author, year, method and findings — best for finding research gaps.

02

Research Design Matrix

Align your own study's research questions, objectives, methods and data sources — best for checking your own study.

03

Synthesis Matrix

Group sources by common theme rather than by article — best for writing your literature review.



HANDS-ON PRACTICE

All exercises, with model answers

Every exercise below matches the hands-on activity on the module's web page. Attempt each one yourself first, then compare against the model answer.

EXERCISE 3.1

~10 min

Knowledge Organisation Prompt Builder

Describe the sources or notes you want to organise. The builder assembles a ready-to-use prompt asking an AI chat tool to structure them into a synthesis matrix, timeline, mind map outline or theoretical framework map.

- ☐ List your sources, notes, or the topic and concepts you want structured.
- ☐ Choose an organisation goal: thematic synthesis matrix, chronological timeline, conceptual mind map, or theoretical framework map.
- ☐ Choose an output format (table, nested outline, or narrative synthesis) and a level of detail.
- ☐ Upload the actual sources alongside the generated prompt so the AI can ground every entry in real material.

✓ WORKED EXAMPLE — GENERATED PROMPT

Sample input — Sources: "Smith 2021, Lee 2022, Tan 2023 on AI in higher education" · Goal: Thematic synthesis matrix · Format: Table · Detail: Brief overview.

Generated prompt: "Act as a research assistant helping me organise source material. Sources, notes or topic to organise: Smith 2021, Lee 2022, Tan 2023 on AI in higher education. Organisation goal: Thematic synthesis matrix. Present the result as a table. Level of detail: Brief overview. Ground every theme, connection and entry in the sources or notes I provided. Do not invent studies, findings or relationships that are not supported by the material. Clearly flag any gaps where more source information is needed."

A good response returns a compact table of themes with each source tagged against the themes it supports, and explicitly flags any theme it could not confidently support from the material provided.

EXERCISE 3.2A

~20 min

Build a Research Matrix — Literature Review Matrix

Organises past studies by author, year, method and findings, one row per source, so you can compare them side by side and spot what the literature has not yet answered. Best for finding research gaps.

- ☐ List 4–6 sources on your topic with author, year and a short abstract for each.
- ☐ Run the ready-to-use prompt below (with your sources pasted in place of the bracketed instruction) in an AI chat tool.
- ☐ Check the resulting table against each source: correct any wrong method, finding, or fabricated detail.
- ☐ Read the 2–3 suggested research gaps and confirm each is actually supported by the pattern in your rows.

AUTHOR, YEAR	METHOD	KEY FINDING	GAP
Smith 2021	Survey, n=210	AI feedback speeds up drafting	Long-term retention not measured
Lee 2022	Case study	Students value fast feedback	Single institution, small sample

✓ READY-TO-USE PROMPT

Act as a research assistant helping me build a literature review matrix.

Sources to include (author, year, and a short description or abstract of each):
[PASTE YOUR SOURCES HERE]

Organise them into a table with these columns: Author & Year, Purpose/Objective, Method & Sample, Key Findings, Limitations, Gap or Question This Raises.

Base every cell only on the sources I provided. Do not invent findings, methods or authors. If a source does not report a column's information, write "Not reported" instead of guessing.

After the table, list 2 to 3 possible research gaps suggested by patterns across the rows.

EXERCISE 3.2B

~20 min

Build a Research Matrix — Research Design Matrix

Aligns your own study’s core components — research questions, objectives, methods and data sources — one row per question, so you can check every part of your design is logically consistent. Best for checking your own study.

- ☐ List your own research question(s), objectives and proposed method(s).
- ☐ Run the ready-to-use prompt below in an AI chat tool.
- ☐ Check whether the AI flags any row where the method does not seem capable of answering the stated question.
- ☐ Revise any misaligned row before treating your design as final.

RESEARCH QUESTION	OBJECTIVE	METHOD	DATA SOURCE
Does AI feedback improve revision quality?	Compare revision quality with and without AI feedback	Quasi-experimental	Pre/post writing samples

✓ READY-TO-USE PROMPT

Act as a research supervisor helping me check the alignment of my study.

My research questions: [LIST YOUR RESEARCH QUESTIONS]
My objectives: [LIST YOUR OBJECTIVES]
My proposed method(s): [LIST YOUR METHODS]

Build a table with one row per research question and these columns: Research Question, Objective, Variables or Constructs, Method, Data Source or Instrument.

Check whether each research question logically aligns with its objective and method. Flag any row where the method does not seem capable of answering the stated question, and explain why.

Build a Research Matrix — Synthesis Matrix

Groups sources by common theme or idea rather than by individual article, one row per theme, so you can see how the literature clusters and where sources agree or disagree. Best for writing your literature review.

- ☐ List 4–6 sources with a short description or key points of each.
- ☐ Run the ready-to-use prompt below in an AI chat tool to identify 3–6 recurring themes.
- ☐ Check that every theme is actually discussed in the sources marked against it — remove any that is not.
- ☐ Note any theme where sources disagree or contradict each other, since this is a strong candidate for your own research gap.

THEME	SUPPORTING SOURCES
AI improves drafting speed	Smith 2021; Lee 2022; Tan 2023
Feedback quality varies by tool	Lee 2022; Wong 2023

✓ READY-TO-USE PROMPT

Act as a research assistant helping me build a thematic synthesis matrix.

Sources (author, year, and a short description or key points of each):
[PASTE YOUR SOURCES HERE]

Identify 3 to 6 recurring themes across these sources. Build a table with themes as rows and sources as columns, marking which sources discuss each theme and a one-line note on their position.

Ground every theme in the sources provided. Do not invent a theme that is not actually discussed. Note any theme where sources disagree or contradict each other.

EXERCISE 3.3

~15 min

Concept mapping from an abstract

Before you let an AI tool draw a mind map for you, try extracting the concepts and relationships yourself from one abstract. Comparing your manual map against the AI's version shows you exactly what a source-grounded tool adds, and what it might miss.

- ☐ Pick one abstract from your shortlist, ideally one with a few distinct concepts and a clear relationship between them.
- ☐ Manually list the key concepts in the abstract, then write how each pair of concepts relates (causes, is a type of, is measured by, contradicts, and so on).
- ☐ Run the concept-extraction prompt below with the same abstract in an AI chat tool.
- ☐ Compare the two: did the AI find a relationship you missed, or add one that is not actually in the text?

CONCEPT	RELATIONSHIP	CONCEPT
AI feedback	reduces	time spent on revision
Time spent on revision	correlates with	student satisfaction

✓ MODEL ANSWER — WORKED EXAMPLE

- Sample abstract (paraphrased): "This study examines how AI-generated feedback affects the writing revision process of undergraduate students. Results show that AI feedback reduces the time spent on revision and is associated with higher reported satisfaction, though causal improvement in writing quality was not established."
- Manual extraction — triples: (AI feedback, reduces, time spent on revision); (time spent on revision, correlates with, student satisfaction); (AI feedback, is not shown to improve, writing quality).
- AI chat tool comparison: the AI correctly extracted the first two triples but initially phrased the third as "AI feedback improves writing quality" — a relationship stronger than the abstract actually supports.
- Correction made: the third triple was revised back to a correlational, not causal, relationship, matching the abstract's actual (cautious) wording — this is exactly the kind of overstatement to catch before using an AI-drawn concept map in your own writing.

WHY USE AI HERE



Traditional vs AI-assisted approach

Task	Traditional Approach	AI-Assisted Approach
Clustering themes	Sort sticky notes or index cards by hand	KO Prompt Builder or Gemini Notebook Studio clusters themes from your sources in seconds
Drawing the mind map	Sketch by hand or manually place nodes one at a time	Markmap or Xmind render a full map instantly from a text outline
Tracing themes to sources	Flip between your notes and the map to check each link	A source-grounded prompt keeps every theme traceable to a named source
Updating as new papers arrive	Redraw or manually re-cluster the whole map	Re-run the same prompt with the new sources added

The trade-off: AI speeds up structuring; you still confirm every theme and connection is grounded in a real source.

COMMON PITFALLS



Things to verify before you trust the output

- ☐ Letting an AI-built matrix or mind map stand without checking every row/node against the actual source material.
- ☐ Choosing the wrong matrix type for the task — e.g., using a Synthesis Matrix when you actually need to check your own study's Research Design alignment.
- ☐ Accepting an AI-suggested relationship between two concepts (e.g., "causes" or "improves") that the source abstract only supports as a correlation.
- ☐ Building a mind map so large it can no longer be explained out loud — a sign it needs pruning or better grouping.
- ☐ Forgetting that a matrix or map is a thinking tool, not a citation — every cell must still trace back to a real source when you write it up.

✧ Key Takeaways

- A synthesis matrix, timeline, mind map or framework map each suit a different organisation goal, so choose based on what you need to show.
- Always ground AI-generated structure in your actual sources; never let it invent themes or connections.
- Mind-mapping tools structure your thinking; Gemini Notebook Studio generates a map directly from verified sources.
- A well-organised literature map becomes the backbone of your literature review section later.

- **What's next:** Have your literature organised into themes? Module 4 helps you turn those findings into clear charts, diagrams and visual research communication.



FOR YOUR NOTES

Session Notes

A large rectangular area with horizontal lines for taking notes.



FOR YOUR NOTES

Session Notes (continued)

A large rectangular area with horizontal lines, intended for taking notes.